Daeyeon Kim

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Research Interests

Integrated photonics for scalable quantum computing
- Integrated Photonics, Quantum Photonics, Nonlinear Photonics

Education

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, Korea
M.S. in Mechanical Engineering — Feb. 2025 – Present
Advisor: Prof. Donguk Nam

Seoul National University (SNU) — Seoul, Korea
B.S. in Mechanical and Aerospace Engineering — Mar. 2019 – Feb. 2025
B.S. in Electrical and Computer Engineering — *(18 Months Military Service)*
Graduated with Honors

Research Experience

Quantum Semiconductor Photonics Integration Lab, KAIST/SNU — Daejeon/Seoul, Korea
Graduate Research Assistant, Advisor: Prof. Donguk Nam — Feb. 2025 – Present

- Developing on-chip intensity difference squeezed-light source via SiN waveguide spontaneous four wave mixing (SFWM). [J1]
- Built a 4 K optical measurement setup for cryogenic device characterization. [J2]
- Established 2D material transfer and optical characterization setups for integrated devices. [J3]
- Modeled and simulated a lithium niobate (LN) based optical phased array (OPA) for chip-scale LiDAR in collaboration with Delta Electronics, Inc.
- Designed microring, directional couplers, edge couplers and grating couplers for two photonic integrated circuit (PIC) tape-outs.

Laser Lab, University of Auckland — Auckland, New Zealand
Visiting Researcher, Advisor: Prof. Stuart Murdoch — Jul. 2026

- Conducted frequency-comb measurements, observed Raman lasing, and characterized critical coupling in integrated photonic devices.
- Collaborated on on-chip squeezed-light source research. [J1]

Nano Energy Transfer and Engineering Lab, SNU — Seoul, Korea
Undergraduate Research Intern, Advisor: Prof. Taeyong Kim — Jun. 2023 – Sep. 2023

- Computed thermal conductivity and phonon lifetime of silicon using density functional theory (DFT).
- Built a time-domain thermoreflectance (TDTR) setup for thermal-property measurement.

Publications

† Equal contribution. * Corresponding authors.

[J1] **D. Kim**, S. Ahn, G. Jeong, J. Im, A. Porritt, S. Murdoch*, D. Nam*. "Cryogenic On-Chip Squeezed Light Generation". In preparation.

[J2] J. Baek†, S. Park†, D. Seo†, H. Kim†, H. Ahn, J. Ha, J. Oh, Y. Choi, S. Lee, **D. Kim**, D. Nam, M. Flavin, B. Jang, H. Yoon, J. Lee, S. Oh*, Y. Song*, D. Kang*. "Ultrafast Optoelectronic Non-

Volatile Memory Enabled by Defect-Mediated Charge Injection in van der Waals Heterostructures". ***Advanced Functional Materials*** (in revision).

[J3] B. Son†, H. Son†, Y. Kim, **D. Kim**, K. Kim, B. Jang, D. Kang*, D. Nam*. "Avalanche-Enhanced Infrared Photodetection via Interlayer Absorption in a WSe_2/MoS_2 Heterostructure". ***Nano Letters*** (2026). DOI: 10.1021/acs.nanolett.6c01893.

Honors and Awards

Government-Funded Scholarship, KAIST — Feb. 2025 – Present
Work-Study Scholarship, SNU — Feb. 2023
Grand Prize, The 8th Army Startup Competition [News 1] [News 2] — Nov. 2022
Merit-Based Scholarship, SNU — Feb. 2021
Bronze Medal, Korean Contest of Mathematics for University Students — Dec. 2020

Teaching Experience

Tutor, SNU
Basic Physics 2 — Sep. 2024 – Dec. 2024
Basic Physics 1, Calculus 1 — Mar. 2024 – Jun. 2024
Foundation of Physics 2 — Sep. 2023 – Dec. 2023

Technical Skills

PIC Design — Lumerical, gdsfactory, KLayout
Fabrication — MOSFET processing (lithography, deposition, etching, metallization)
Programming — Python, MATLAB, C/C++
Other Tools — Quantum ESPRESSO, LTspice, SOLIDWORKS

Extracurricular Activities

Military Service, Republic of Korea Army — Aug. 2021 – Feb. 2023
Satellite Communications Operator, Sergeant
- Operated satellite and radio communication systems and secure voice equipment.

Run To You, SNU Society of Automobile Engineers Team — Mar. 2019 – Feb. 2020
EV Team Technician
- Designed and assembled the electric vehicle (EV) drivetrain system.

Science Outreach Volunteer, Pyeongchang Elementary School — Aug. 2019
Science Experiment Mentor
- Led hands-on science experiments and taught fundamental scientific concepts to elementary school students.